

Soheil SARHANG ZADEH

Control theory and mechatronics student

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4th year electrical engineering student at Ecole Centrale de Nantes with an international background (France, UK, Singapore), specializing in advanced control systems. Seeking a 4-month R&D internship starting April 2027.

Technical projects

Implementing FOC for 3D printers

| 2026

- Integrated SimpleFOC SDK on STM32 with A4950 driver and magnetic encoder into Klipper 3D printer firmware.
- Parsed and converted Klipper motion planning output to position trajectories, validated on live print data.
- Hardware integration in progress.

Automatic Inspection Machine for PCBs

| 2026 Internship

- Converted a 3D printer chassis into an automated PCB inspection gantry controlled by a Raspberry Pi.
- Developed motor control loops and camera trigger synchronization in C++/Python; collaborated with CAD and computer vision teams.

Solar panel MPPT controller

| 2024

- Designed and built a Buck-Boost converter controlled by Arduino to maximize solar panel power extraction.
- Developed control algorithms in Matlab and successfully tested them with a PV panel. Validated experimentally against baseline

Amplifier PCB

| 2024

- Designed a PCB in KiCad around a power amplifier IC, handling component selection and schematic design.
- Hand-soldered SMD and through-hole components and validated the prototype experimentally.

Work experience

ICRF — RF/Microwave Design and Manufacturing Internship,

| Caen, France 2026

- Monitored and operated semi-automated SMT line for high-frequency RF cards, covering pick-and-place machines, reflow oven profiling, and automated optical inspection.
- Assembly & QA: Assembled custom high-frequency coaxial cable assemblies and performed electrical QA testing to verify RF performance and manufacturing consistency.

Skills and languages

- Software : Matlab/Simulink, KiCad, LTSpice, Git, Linux
- Programming : Python, Cpp
- Soldering, lab equipment (oscilloscopes, power analysers)
- Fluent in English and French, B1 in German

Education

École Centrale de Nantes - Nantes, France

| 2026 – 2029

- Control Theory: Advanced & Predictive Control (MPC, Sliding Mode), Methodology of Linear Control (State-space Observers, Kalman filter).
- Hardware & Power: Energy Conversion Chains (Motors & Power Electronics), Digital Design on FPGA (VHDL)
- Embedded Computing (Bare-metal microcontrollers, RTOS).

Lycée Branly Lyon

| Prep school 2023 – 2026